

Lecture 3: Regression Modeling for Areal Data

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Spatial Regression Models

Y_i is a response variable at spatial unit i . We wish to model:

$$Y_i = \sum_{k=1}^p \beta_k x_{ik} + \epsilon_i = \mathbf{x}_i' \boldsymbol{\beta} + \epsilon_i$$

where $\mathbf{x}_i$ is a row vector of covariate and ϵ_i is the residual error.

Goals: Estimate $\boldsymbol{\beta}$ accounting for spatial autocorrelation in ϵ_i .

Methods:

1. Conditional Autoregressive Model (CAR) - the focus here
2. Simultaneous Autoregressive Model (SAR)

For now, we will focus on the case where Y_i can be modeled as a **Gaussian** outcome.

Conditional Autoregressive Models

Consider the **conditional** specification of the random variable Y_i given all the other spatial units as:

$$[Y_i | Y_j : j \neq i] \sim N \left(\rho \sum_{j=1}^n b_{ij} Y_j, \tau_i^2 \right) .$$

The above model indicates that the *full conditional* distribution of Y_i :

- ▶ is Normal;
- ▶ has a mean proportional (via $\rho > 0$) to a linear combination of all other random variables. The weights are defined by b_{ij} with $b_{ii} = 0$;
- ▶ has a location-specific conditional variance τ_i^2 .

What is the induced *joint* distribution? What conditions on ρ , b_{ij} , and τ_i^2 do we need to ensure a valid joint distribution?

Conditional Autoregressive Model

To get a valid joint distribution, the conditional distribution of each Y_i must satisfy

$$[Y_i | Y_j : j \neq i] \sim N \left(\rho \frac{1}{W_{i+}} \sum_{j=1}^n W_{ij} Y_j, \frac{\tau^2}{W_{i+}} \right) .$$

We typically assume W_{ij} to be binary (indicator for being neighbors).

Then

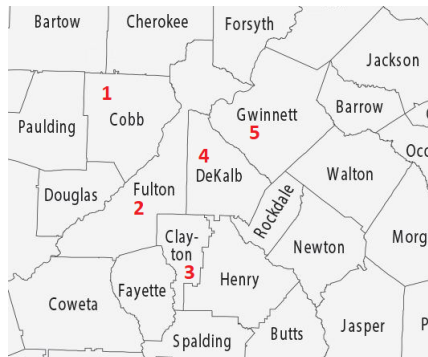
- ▶ The conditional mean of Y_i is the **weighted average** of all neighbors.
- ▶ The conditional variance is **proportional to the number** of neighbors.

The induced joint distribution is multivariate Normal!

$$[Y_1, Y_2, \dots, Y_n] \sim N(\mathbf{0}, \tau^2(\mathbf{D}_w - \rho \mathbf{W})^{-1})$$

where $\mathbf{D}_w = \text{diag}(W_{1+}, W_{2+}, \dots, W_{n+})$ and $\mathbf{W}$ is the adjacency matrix.

An Example: 5-county Atlanta



$$W = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$W_{1+} = 1 \quad W_{2+} = 3 \quad W_{3+} = 2 \quad W_{4+} = 3 \quad W_{5+} = 1$$

An Example: 5-county Atlanta

$$[Y_i | Y_j : j \neq i] \sim N \left(\rho \frac{1}{W_{i+}} \sum_{j=1}^n W_{ij} Y_j, \frac{\tau^2}{W_{i+}} \right) .$$

Conditional Distribution for Y_1 , Y_2 and Y_3 :

$$Y_1 | Y_2, Y_3, Y_4, Y_5 \sim N(\rho Y_2, \tau^2)$$

$$Y_2 | Y_1, Y_3, Y_4, Y_5 \sim N(\rho(Y_1 + Y_3 + Y_4 + Y_5)/4, \tau^2/4)$$

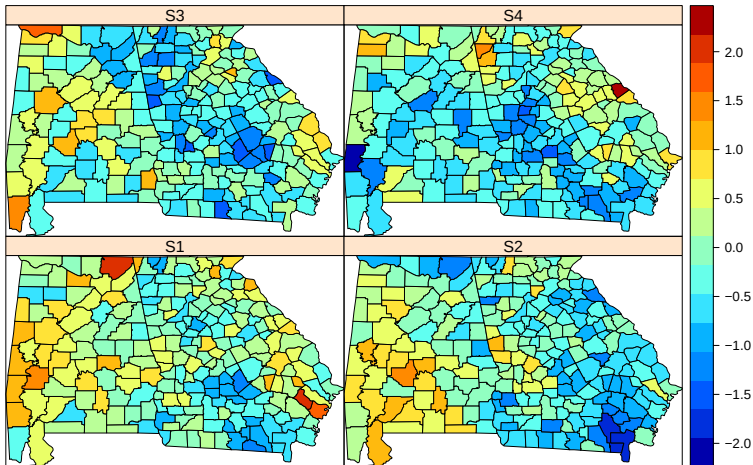
$$Y_3 | Y_1, Y_2, Y_4, Y_5 \sim N(\rho(Y_2 + Y_4)/2, \tau^2/2)$$

Joint Distribution for Y_1 , Y_2 and Y_3 :

$$\begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \\ Y_4 \\ Y_5 \end{bmatrix} \sim N \left(\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \tau^2 \begin{bmatrix} 1 & -\rho & 0 & 0 & 0 \\ -\rho & 3 & -\rho & -\rho & 0 \\ 0 & -\rho & 2 & -\rho & 0 \\ 0 & -\rho & -\rho & 3 & -\rho \\ 0 & 0 & 0 & -\rho & 1 \end{bmatrix}^{-1} \right)$$

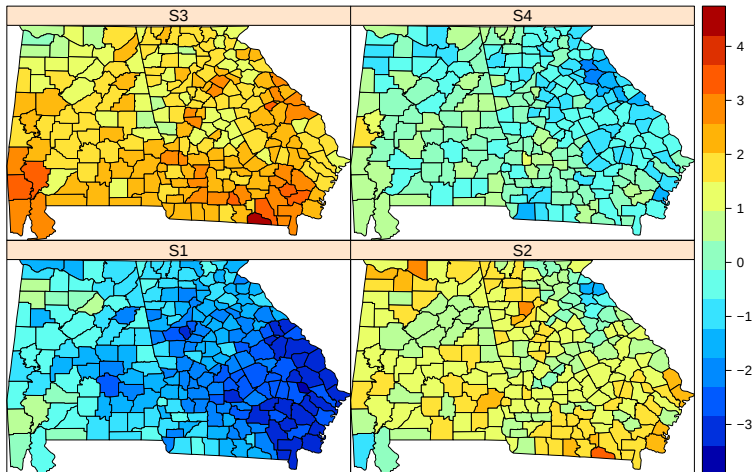
Example Simulations

CAR ($\rho = 0.99, \tau^2 = 1$)



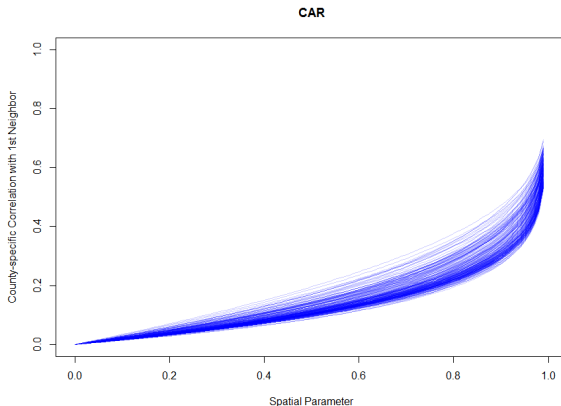
Example Simulations

CAR ($\rho = 0.9999, \tau^2 = 1$)



Impacts of ρ on Average 1st Neighbor Correlation

Below we plot the induced average correlation between a county and its neighbors as a function of ρ .



Note that even with $\rho = 0.99$, the correlation is still < 0.8 !

CAR Parameters

$$[Y_1, Y_2, \dots, Y_n] \sim N(\mathbf{0}, \tau^2(\mathbf{D}_w - \rho\mathbf{W})^{-1}) .$$

Several important properties of the covariance matrix should be noted.

- ▶ τ^2 cannot be interpreted as the variance at each location. In $(\mathbf{D}_w - \rho\mathbf{W})^{-1}$, the diagonal elements (variance) are dependent on ρ and $\mathbf{W}$.
- ▶ $\rho = 0$ implies that Y_i are independent; but the marginal variance $(\tau^2\mathbf{D}_w^{-1})$ is proportional to the number of neighbors. This is not a desirable property for independent data!
- ▶ If $\rho = 1$, then the joint distribution is invalid because $\mathbf{D}_w - \mathbf{W}$ is not invertible (it is of rank $n - 1$). Therefore, $\rho = 1$ cannot be used to model observed data; but can be used as prior distribution for random effects (particularly in Bayesian analysis later). When $\rho = 1$, we often refer to it as the **intrinsic** or **improper** CAR model.

CAR Model Example

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i \quad \epsilon \sim N(0, \tau^2(\mathbf{D}_w - \rho \mathbf{W})^{-1})$$

- ▶ y_i = chlamydia incidence per 10,000.
- ▶ x_i = household income (centered and divided by 1,000).

		Estimate (Standard Error)				
	W	β_0	β_1	λ	τ^2	AIC
LM	NA	46 (1.6)	-1.31 (0.18)	NA	24	2089
CAR	1st-order	42 (3.5)	-1.33 (0.22)	0.15	21	2045
CAR	2nd-order	38 (3.9)	-1.14 (0.21)	0.08	21	2051

- ▶ Household income and chlamydia incidence are negatively associated.
- ▶ Note that standard errors for regression coefficients increase when accounting for spatial correlation.
- ▶ Using AIC as a model selection criterion, CARS with 1st-order adjacency is preferred.

Advantages of CAR

The spatial dependence structure is contained entirely in $(\mathbf{D}_w - \rho \mathbf{W})^{-1}$. The matrix $\mathbf{Q} = (\mathbf{D}_w - \rho \mathbf{W})$ is known as the **precision matrix**. The CAR model falls under a class of Markov network or undirected graphical model known as **Gaussian Markov random field** (GMRF).

- ▶ GMRF theory says that

$$\mathbf{Q}_{ij} = 0 \quad \Leftrightarrow \quad Y_i \perp\!\!\!\perp Y_j \mid \mathbf{Y}_k, k \neq i, j$$

Therefore, if the proximity matrix $\mathbf{W}$ is zero between Y_i and Y_j , it implies that they are conditionally independent. It is often easier to describe conditional dependence than joint dependence.

- ▶ In practice, $\mathbf{W}$ contains a large number of zeros. Computation associated with the normal density can be done efficiently with $\mathbf{Q}^{-1}$ using sparse matrix algorithms (e.g. $n > 20,000$).

Modeling Areal Count Data and Disease Mapping

Setup:

- ▶ Counts of disease incidence or prevalence aggregated over some contiguous administrative spatial areal units.
- ▶ Each count is associated with certain covariates (e.g. at-risk population size) at the same spatial unit.

Goals:

1. **Regression:** explain spatial variation in diseases rates as a function of covariates.
2. **Smoothing:** provide location-specific estimates that have better precision.

Smoothing = Borrowing Information

Poisson Log-Linear Model

For spatial unit $i = 1, \dots, n$, assume the following Poisson log-linear model

$$Y_i \sim \text{Poisson}(\lambda_i)$$
$$\log(\lambda_i) = \mathbf{X}_i' \boldsymbol{\beta}.$$

- ▶ Y_i is the observed case count.
- ▶ λ_i is the location-specific Poisson mean with fixed-effect $\boldsymbol{\beta}$ and covariates $\mathbf{X}$.

Problems with this model:

1. λ_i is completely determined by the covariates and $\boldsymbol{\beta}$ is shared across ALL spatial units.
2. Does not account for spatial variability/dependence not explained by $\mathbf{X}$ (often leading to over/under-dispersion).

Poisson Log-Linear Model with Random Effects

First, we can include a location-specific effect (u_i) in the mean structure:

$$Y_i \sim \text{Poisson}(\lambda_i)$$
$$\log(\lambda_i) = \mathbf{X}_i' \boldsymbol{\beta} + u_i .$$

- ▶ u_i can be interpreted as the location-specific log baseline mean count.
- ▶ u_i is difficult (almost impossible) to estimate well because we only have one observation at each location!

Under a hierarchical modeling framework, we treat u_i as random effects:

$$u_i \stackrel{iid}{\sim} N(0, \sigma^2) .$$

This allows shrinkage, information-borrowing, or penalization on u_i based on the value of σ^2 . Smaller σ^2 results in higher shrinkage towards 0.

A Note on Offset

In the Poisson mean, we often include an offset $\log P_i$:

$$\log(\lambda_i) = \log P_i + \mathbf{X}_i' \boldsymbol{\beta} + u_i .$$

This implies the Poisson model

$$Y_i \sim \text{Poisson} (P_i \exp\{\mathbf{X}_i' \boldsymbol{\beta} + u_i\}) .$$

Two typical choices for P_i :

1. at-risk population size; or
2. expected disease count.

With offsets, $\exp\{\mathbf{X}_i' \boldsymbol{\beta} + u_i\}$ can be interpreted as

1. prevalence/incidence **rates** for the at-risk population; or
2. **relative deviation** from the expected counts.

Spatial Random Effects

$$\log(\lambda_i) = \log P_i + \mathbf{X}_i' \boldsymbol{\beta} + u_i .$$

Often $\mathbf{X}$ is not able to remove all the spatial dependence in u_i . So we can also consider u_i 's as spatially-dependent random effects.

Let $\mathbf{u}' = (u_1, \dots, u_n)$, we assume the joint distribution is Normal with covariance $\boldsymbol{\Sigma}$:

$$\mathbf{u} \sim N(\mathbf{0}, \boldsymbol{\Sigma}) .$$

For areal data, the most commonly used spatial random effect specification is the **conditional autoregressive** (CAR) model.

Let $\mathbf{W}$ denote the symmetric adjacency matrix. The CAR model is given by two parameters ρ and τ^2 :

$$\mathbf{u} \sim N(\mathbf{0}, \tau^2 [\mathbf{D} - \rho \mathbf{W}]^{-1})$$

is equivalent to a conditional specification, for $i = 1, \dots, n$

$$u_i | u_{j \neq i} \sim N \left(\rho \frac{\sum_{j \neq i} W_{ij} u_j}{D_{ii}}, \frac{\tau^2}{D_{ii}} \right)$$

Proper versus Improper CAR

An **improper** or **intrinsic** CAR is a special case where $\rho = 1$

$$u_i | u_{j \neq i} \sim N \left(\frac{\sum_{j \neq i} W_{ij} u_j}{D_{ii}}, \frac{\tau^2}{D_{ii}} \right) .$$

However, this does not correspond to a valid joint multivariate Normal distribution.

We can see that if we add a constant k to all the u_i , the above distribution still holds.

$$u_i + k | u_{j \neq i} + k \sim N \left(\frac{\sum_{j \neq i} W_{ij} (u_j + k)}{D_{ii}}, \frac{\tau^2}{D_{ii}} \right) .$$

To get around this, we often impose a constraint for improper CAR:

$$\sum_{i=1}^n u_i = 0 .$$

Improper CAR in Bayesian Analysis

The improper CAR is usually used under a Bayesian framework, which makes inference on the posterior distribution of u_i :

$$[\mathbf{u} | \mathbf{Y}] = \frac{[\mathbf{Y}, \mathbf{u}]}{[\mathbf{y}]} = \frac{[\mathbf{Y} | \mathbf{u}] \times [\mathbf{u}]}{[\mathbf{y}]} \propto [\mathbf{Y} | \mathbf{u}] \times [\mathbf{u}].$$

So we can use CAR as a prior for $[\mathbf{u}]$ and the posterior distribution will be valid as long as we have data $\mathbf{Y}$.

Note: The reason we can do this here is because

- ▶ We assume u_i is a random effect.
- ▶ **Conditioned** on u_i , the stochastic nature of Y_i is given by the Poisson likelihood - NOT by the CAR prior.

Leroux's Proper CAR

Recall that one disadvantage of the proper CAR model

$$\mathbf{u} \sim N(\mathbf{0}, \tau^2[\mathbf{D} - \rho\mathbf{W}]^{-1})$$

is that it does not reduce to an independent model with $\rho = 0$.

Another commonly used structure is the Leroux's form:

$$\mathbf{u} \sim N(\mathbf{0}, \tau^2[(1 - \rho)\mathbf{I} + \rho(\mathbf{D} - \mathbf{W})]^{-1})$$

The above reduces to iCAR when $\rho = 1$ and an exchangeable model when $\rho = 0$. One challenge is that the conditional distribution becomes a bit less intuitive:

$$u_i | u_{j \neq i} \sim N\left(\frac{\rho \sum_{j \neq i} W_{ij} u_j}{1 - \rho + \rho D_{ii}}, \frac{\tau^2}{1 - \rho + \rho D_{ii}}\right).$$

The above conditional mean/variance can be seen as weighted mean between the iCAR and exchangeable model with weight (for iCAR) $= \rho D_{ii} / (1 - \rho + \rho D_{ii})$.

Convolution Model

A convolution model (aka Besag-York-Mollie, BYM) includes both **spatially-dependent** and **unstructured/exchangeable** random effects.

$$\log(\lambda_i) = \log P_i + \mathbf{X}_i' \boldsymbol{\beta} + u_i + v_i .$$

$$\mathbf{u} \sim N(\mathbf{0}, \tau^2 [\mathbf{D} - \mathbf{W}]^{-1})$$

$$v_i \stackrel{iid}{\sim} N(0, \sigma^2) .$$

Challenge!!! We are asking our model to partition a location-specific random effect into a spatial and a non-spatial component. With only one Y_i at each location, this is a very difficult task. In this setting, our main inferential interest may be on $u_i + v_i$.

Chlamydia Application

First consider models without covariates:

$$Y_i \sim \text{Poisson}(\lambda_i) \quad \log(\lambda_i) = \log P_i + \beta_0 + u_i.$$

where Y_i is the chlamydia counts and P_i is the population size.

Model 1: Crude Estimate (MLE) :

$$\hat{\beta}_0 + \hat{u}_i = \log(Y_i/P_i).$$

Model 2: Exchangeable (independent)

$$u_i \sim N(0, \sigma^2)$$

Model 3: Only Spatial (improper CAR)

$$u_i \sim iCAR(\tau^2)$$

Model 4: Only Spatial (proper Leroux)

$$u_i \sim pCAR(\tau^2)$$

Model 5: Convolution (iCAR + exchangeable)

$$u_i = \gamma_i + \theta_i$$

$$\gamma_i \sim iCAR(\tau^2) \quad \theta_i \sim N(0, \sigma^2)$$

Bayesian Inference

We often frame estimation in disease mapping under a Bayesian framework to **get uncertainty estimates for random effects**.

Typical priors:

- ▶ $\beta_0 \sim \text{Normal}(0, \sigma_\beta^2)$ where σ_β^2 can be 0 (improper flat prior) or be very large (e.g., 100^2).
- ▶ Variance components $\tau \sim \text{Inv-Gamma}(a, b)$ with small a and b (e.g. 0.0001).
- ▶ $\text{Logit}(\rho) \sim \text{Normal}(0, \sigma_\rho^2)$.

Estimation can be done using MCMC-based methods (R Package CARBayes) or approximation methods (INLA).

Model Estimates

Posterior Median and 95% Posterior Interval

	β_0	τ^2	σ^2
Exch.	-5.57 (-5.66, -5.57)		0.42 (0.35, 0.51)
iCAR	-5.57 (-5.58, -5.55)	1.21 (1.13, 1.48)	
pCAR	-5.57 (-5.95, -5.19)	1.08 (0.85, 1.37)	
Conv.	-5.57 (-5.62, -5.52)	0.60 (0.34, 1.03)	0.12 (0.06, 0.19)

For pCAR, estimate of ρ is 0.81 (0.59, 0.95).

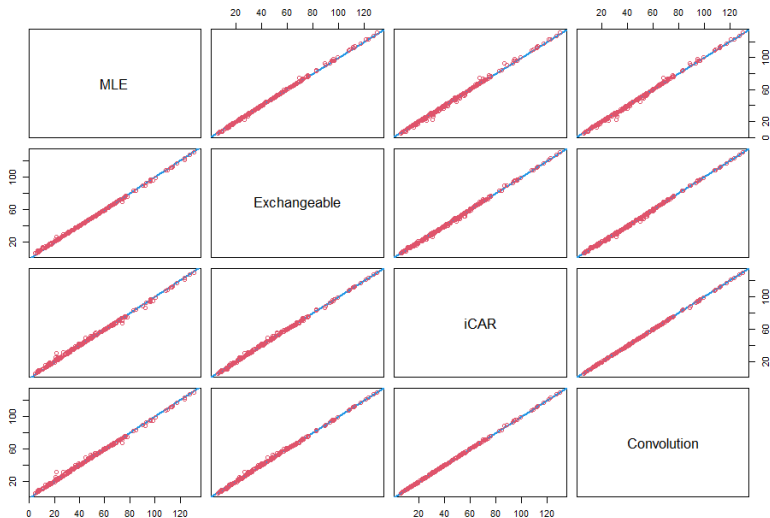
Model Comparison

	DIC	pD	WAIC	pWAIC
Exch.	1926	218	1873	118
iCAR	1917	211	1864	114
pCAR	1920	213	1870	117
Conv.	1919	213	1868	116

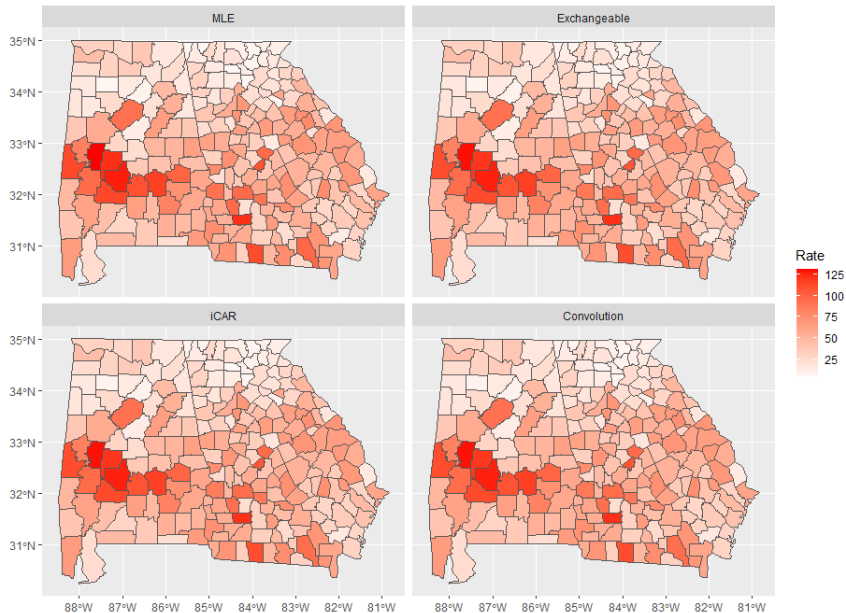
DIC/WAIC prefers the spatial models, which also gives the smallest number of effective degrees of freedom (pD).

Estimated Relative Risk (per 10,000)

County-specific estimates are very similar.

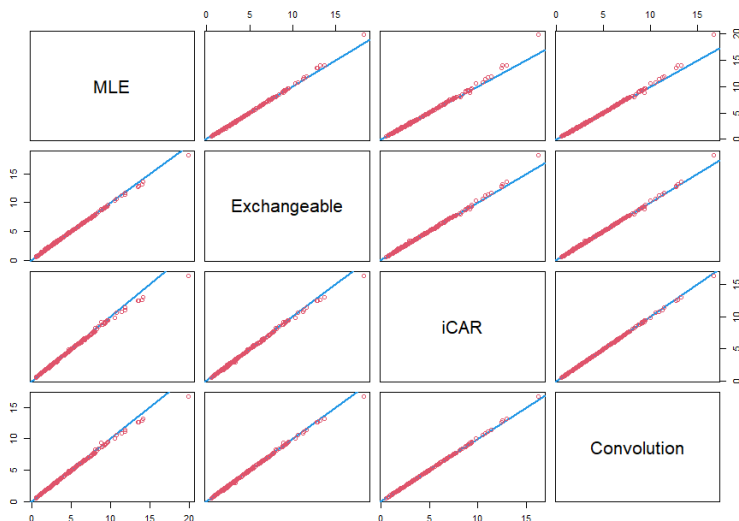


Estimated Relative Risk (per 10,000)



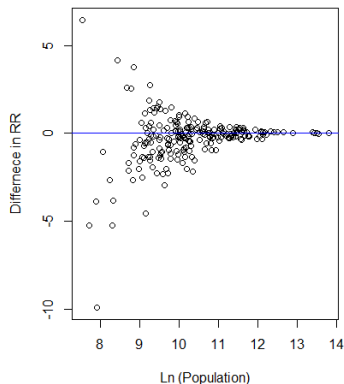
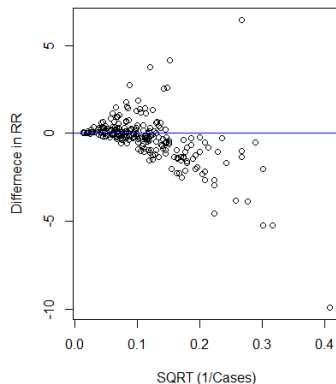
Comparison of County-Specific Standard Error

Larger SE's from crude (MLE) are reduced in random effect models.



Difference in Estimated Log Rate ($\hat{\beta}_0 + \hat{u}_i$)

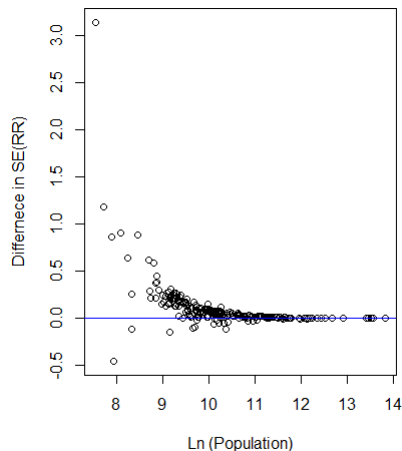
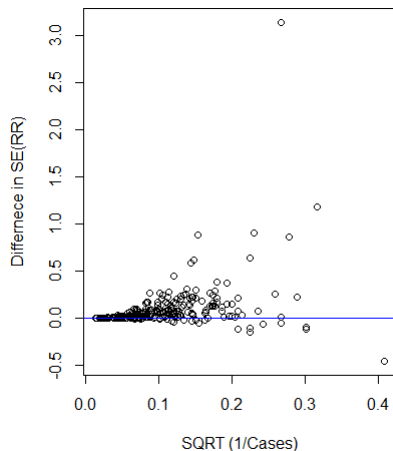
MLE (Model 1) - Convolution (Model 5)



Larger differences observed when the **the case number is larger** or **population is smaller**.

Difference in Standard Error

MLE (Model 1) - Convolution (Model 5)



Crude estimates have larger standard errors when the **the case number is larger** or **population is smaller**.

Convolution Model with Covariate

$$Y_i \sim \text{Poisson}(\lambda_i)$$

$$\log(\lambda_i) = \log P_i + \beta_0 + \beta_1 \text{Income}_i + u_i + v_i$$

$$u_i \sim \text{ICAR}(\tau^2) \quad v_i \sim N(0, \sigma^2)$$

	Posterior Mean	95% Post Int
β_0	-5.56	(-5.60, -5.53)
$\beta_1 (\times 1,000)$	-0.034	(-0.043, -0.024)
τ^2	0.69	(0.41, 1.15)
σ^2	0.06	(0.02, 0.11)

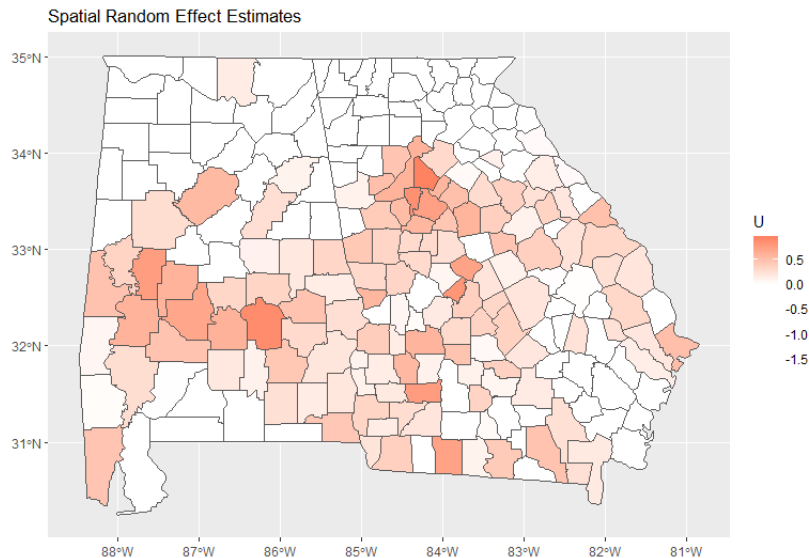
- ▶ σ^2 decreased with the inclusion of income in the model.
- ▶ We found that a \$1,000 increase in county-level median household income was associated with a 3.4% (95% PI: 2.4%, 4.3%) decrease in chlamydia incidence, **controlling for county-level baseline heterogeneity**.

Comparison with Other Models

	Estimates of $\beta_1 (\times 1,000)$		
	Posterior Mean	SE	95% PI/CI
Spatial Convolution	-0.034	0.0049	(-0.043, -0.024)
Poisson Regression	-0.023	0.0004	(-0.024, -0.022)
Quasi-poisson Regression	-0.023	0.0033	(-0.016, -0.029)

- ▶ A standard Poisson regression under-estimates the uncertainty considerably.
- ▶ Using quasi-Poisson to account for over-dispersion increases confidence interval length (by a lot).
- ▶ The point estimates differ as well.

Posterior Mean of Spatial Residuals u_i



Posterior Mean of Independent Residuals v_i

